

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV Examination, 2022-23

Paper: BCH-224: Business Mathematics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Attempt all the questions. All questions carry equal marks (14 marks each).***Question 1.**

- (a) The p^{th} and q^{th} terms of an A.P. are c and d respectively. Find the r^{th} term. [7]
- (b) There are n arithmetic means between 1 and 51. If the ratio between the 4^{th} and 7^{th} means is $3 : 5$, find n . [7]

OR

- (a) Show that the sum of the 5^{th} term from the beginning and the 5^{th} term from the end of an A.P. is equal to the sum of the first and last terms. [7]
- (b) The sum of n terms of an A.P. is 40, the common difference is 2, and the last term is 13. Find the number of terms. [7]

Question 2.

- (a) The $(p + q)^{\text{th}}$ term of a G.P. is m and the $(p - q)^{\text{th}}$ term is n . Find the p^{th} term. [7]
- (b) The arithmetic mean between two numbers is 10 and their geometric mean is 8. Find the numbers. [7]

OR

- (a) Find the G.P. sequence whose 5^{th} and 8^{th} terms are 1 and 8. [7]
- (b) Find the sum of n terms of the series: [7]

$$7 + 77 + 777 + 7777 + \dots$$

Question 3.

- (a) Prove that: [7]

$${}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1} = {}^nP_r$$

- (b) How many numbers between 100 and 10,000 can be formed with the digits 1, 2, 3, 0, 4, 5, 6, 7? [7]

OR

- (a) How many triangles can be formed by joining the angular points of a decagon? How many diagonals does it have? [7]
- (b) If $m = {}^nC_2$, prove that ${}^mC_2 = 3 \cdot {}^{n+1}C_4$. [7]

Question 4.

- (a) Find the 9th term of $(\sqrt{x} - \sqrt{y})^{16}$. [7]
- (b) In the expansion of $(1+x)^{24}$, if the coefficients of the r^{th} and $(r+4)^{\text{th}}$ terms are equal, find r . [7]

OR

- (a) Find the coefficient of x^{32} in the expansion of: [7]
- $$\left(x^4 - \frac{1}{x^3}\right)^{15}$$
- (b) Find the middle term in the expansion of $(3x - y)^8$. [7]

Question 5.

- (a) If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, prove that: [5]

$$f(a) + f(b) = f\left(\frac{a+b}{1+ab}\right)$$

- (b) Evaluate: [4]

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$$

- (c) Differentiate with respect to x : [5]

$$\frac{a^2 + x^2}{\sqrt{a^2 - x^2}}$$

OR

Evaluate the following integrals with respect to x : [5+4+5]

(a) $\int \frac{\sqrt{8+3\sqrt{x}}}{5x\sqrt{x}} dx$

(b) $\int (x\sqrt{x} + 3) dx$

(c) $\int x^n \log x dx$

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